

Chapter 4 Cox Survival Regression

The following code fits the Proportional Cox Hazard model.

```
# Proportional Hazards Model #  
recid.ph <- coxph(Surv(week, arrest) ~ fin + age + wexp + mar + paro + prio, data  
summary(recid.ph)
```

```
## Call:
## coxph(formula = Surv(week, arrest) ~ fin + age + wexp + mar +
##      paro + prio, data = recid)
##
##      n= 432, number of events= 114
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## fin   -0.36554    0.69382  0.19090 -1.915  0.05552 .
## age   -0.05633    0.94523  0.02189 -2.573  0.01007 *
## wexp  -0.15699    0.85471  0.21208 -0.740  0.45916
## mar   -0.47130    0.62419  0.38027 -1.239  0.21520
## paro  -0.07792    0.92504  0.19530 -0.399  0.68991
## prio   0.08966    1.09380  0.02871  3.123  0.00179 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## fin      0.6938      1.4413      0.4773      1.0087
## age      0.9452      1.0579      0.9055      0.9867
## wexp     0.8547      1.1700      0.5640      1.2952
## mar      0.6242      1.6021      0.2962      1.3152
## paro     0.9250      1.0810      0.6308      1.3564
## prio     1.0938      0.9142      1.0340      1.1571
##
## Concordance= 0.639 (se = 0.027 )
## Likelihood ratio test= 32.14 on 6 df,  p=2e-05
## Wald test              = 30.79 on 6 df,  p=3e-05
## Score (logrank) test = 32.28 on 6 df,  p=1e-05
```

Parameter Interpretation

```
recid.ph2 <- coxph(Surv(week, arrest) ~ fin + age + prio, data = recid)
summary(recid.ph2)
```

```
## Call:
```

```
## coxph(formula = Surv(week, arrest) ~ fin + age + prio, data = recid)
```

```
##
```

```
## n= 432, number of events= 114
```

```
##
```

```
##          coef exp(coef) se(coef)      z Pr(>|z|)
```

```
## fin  -0.34695    0.70684  0.19025 -1.824 0.068197 .
```

```
## age  -0.06711    0.93510  0.02085 -3.218 0.001289 **
```

```
## prio  0.09689    1.10174  0.02725  3.555 0.000378 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
##          exp(coef) exp(-coef) lower .95 upper .95
```

```
## fin      0.7068      1.4148      0.4868      1.0263
```

```
## age      0.9351      1.0694      0.8977      0.9741
```

```
## prio     1.1017      0.9077      1.0444      1.1622
```

```
##
```

```
## Concordance= 0.63 (se = 0.027 )
```

```
## Likelihood ratio test= 29.05 on 3 df, p=2e-06
```

```
## Wald test              = 27.94 on 3 df, p=4e-06
```

```
## Score (logrank) test = 29.03 on 3 df, p=2e-06
```

```
(exp(coef(recid.ph2))-1)*100
```

```
##          fin          age          prio
## -29.31625  -6.49033  10.17427
```

```
# Automatic Selection Techniques #
```

```
full.model <- coxph(Surv(week, arrest) ~ fin + age + wexp + mar + paro + prio,
                    data = recid)
```

```
empty.model <- coxph(Surv(week, arrest) ~ 1, data = recid)
```

```
step.model <- step(empty.model,
                   scope = list(lower=formula(empty.model),
                                upper=formula(full.model)),
                   direction = "both")
```

```
## Start:  AIC=1350.76
```

```
## Surv(week, arrest) ~ 1
```

```
##
```

```
##          Df          AIC
```

```
## + age    1 1337.5
```

```
## + prio   1 1340.8
```

```
## + wexp   1 1343.1
```

```
## + mar    1 1348.1
```

```
## + fin    1 1348.9
```

```
## <none>    1350.8
```

```
## + paro   1 1352.4
```

```
##
```

```
## Step:  AIC=1337.49
```

```
## Surv(week, arrest) ~ age
```

```
##
```

```
##           Df      AIC
## + prio    1 1329.1
## + wexp     1 1336.4
## + fin      1 1336.5
## <none>      1337.5
## + mar      1 1337.5
## + paro     1 1338.6
## - age      1 1350.8
##
## Step:   AIC=1329.08
## Surv(week, arrest) ~ age + prio
##
##           Df      AIC
## + fin      1 1327.7
## + mar      1 1329.0
## <none>      1329.1
## + wexp     1 1330.0
## + paro     1 1330.9
## - prio     1 1337.5
## - age      1 1340.8
##
## Step:   AIC=1327.71
## Surv(week, arrest) ~ age + prio + fin
##
##           Df      AIC
## + mar      1 1327.3
## <none>      1327.7
## + wexp     1 1328.6
## - fin      1 1329.1
## + paro     1 1329.4
## - prio     1 1336.5
```

```
## - age    1 1338.3
##
## Step:   AIC=1327.35
## Surv(week, arrest) ~ age + prio + fin + mar
##
##           Df    AIC
## <none>      1327.3
## - mar      1 1327.7
## + wexp     1 1328.8
## - fin      1 1329.0
## + paro     1 1329.2
## - age      1 1335.4
## - prio     1 1336.2
```

```
summary(step.model)
```

```
## Call:
## coxph(formula = Surv(week, arrest) ~ age + prio + fin + mar,
##       data = recid)
##
##    n= 432, number of events= 114
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## age  -0.06042    0.94137  0.02085 -2.897  0.00376 **
## prio   0.09751    1.10243  0.02722  3.583  0.00034 ***
## fin  -0.36020    0.69753  0.19049 -1.891  0.05864 .
## mar  -0.53312    0.58677  0.37276 -1.430  0.15266
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## age      0.9414      1.0623      0.9037      0.9806
## prio      1.1024      0.9071      1.0452      1.1628
## fin      0.6975      1.4336      0.4802      1.0132
## mar      0.5868      1.7042      0.2826      1.2183
##
## Concordance= 0.633 (se = 0.027 )
## Likelihood ratio test= 31.41 on 4 df,  p=3e-06
## Wald test              = 29.98 on 4 df,  p=5e-06
## Score (logrank) test = 31.25 on 4 df,  p=3e-06

full.model <- coxph(Surv(week, arrest) ~ fin + age + wexp + mar + paro + prio,
                    data = recid)

back.model <- step(full.model, direction = "backward")
```

```
## Start:  AIC=1330.62
## Surv(week, arrest) ~ fin + age + wexp + mar + paro + prio
##
##          Df      AIC
## - paro    1 1328.8
## - wexp     1 1329.2
## - mar      1 1330.3
## <none>      1330.6
## - fin      1 1332.3
## - age      1 1336.3
## - prio     1 1337.2
##
## Step:  AIC=1328.78
## Surv(week, arrest) ~ fin + age + wexp + mar + prio
##
##          Df      AIC
## - wexp     1 1327.3
## - mar      1 1328.6
## <none>      1328.8
## - fin      1 1330.4
## - age      1 1334.3
## - prio     1 1335.8
##
## Step:  AIC=1327.35
## Surv(week, arrest) ~ fin + age + mar + prio
##
##          Df      AIC
## <none>      1327.3
## - mar      1 1327.7
## - fin      1 1329.0
## - age      1 1335.4
```

```
## - prio 1 1336.2
```

```
summary(back.model)
```

```
## Call:
```

```
## coxph(formula = Surv(week, arrest) ~ fin + age + mar + prio,  
##       data = recid)
```

```
##
```

```
## n= 432, number of events= 114
```

```
##
```

```
##          coef exp(coef) se(coef)      z Pr(>|z|)  
## fin  -0.36020   0.69753  0.19049 -1.891  0.05864 .  
## age  -0.06042   0.94137  0.02085 -2.897  0.00376 **  
## mar  -0.53312   0.58677  0.37276 -1.430  0.15266  
## prio  0.09751   1.10243  0.02722  3.583  0.00034 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
##          exp(coef) exp(-coef) lower .95 upper .95  
## fin          0.6975      1.4336    0.4802    1.0132  
## age          0.9414      1.0623    0.9037    0.9806  
## mar          0.5868      1.7042    0.2826    1.2183  
## prio         1.1024      0.9071    1.0452    1.1628
```

```
##
```

```
## Concordance= 0.633 (se = 0.027 )
```

```
## Likelihood ratio test= 31.41 on 4 df, p=3e-06
```

```
## Wald test = 29.98 on 4 df, p=5e-06
```

```
## Score (logrank) test = 31.25 on 4 df, p=3e-06
```

```
full.model <- coxph(Surv(week, arrest) ~ fin + age + wexp + mar + paro + prio,
  data = recid)
```

```
empty.model <- coxph(Surv(week, arrest) ~ 1, data = recid)
```

```
for.model <- step(empty.model,
  scope = list(lower=formula(empty.model),
    upper=formula(full.model)),
  direction = "forward")
```

```
## Start: AIC=1350.76
```

```
## Surv(week, arrest) ~ 1
```

```
##
```

```
##      Df    AIC
```

```
## + age  1 1337.5
```

```
## + prio  1 1340.8
```

```
## + wexp  1 1343.1
```

```
## + mar   1 1348.1
```

```
## + fin   1 1348.9
```

```
## <none>   1350.8
```

```
## + paro  1 1352.4
```

```
##
```

```
## Step: AIC=1337.49
```

```
## Surv(week, arrest) ~ age
```

```
##
```

```
##      Df    AIC
```

```
## + prio  1 1329.1
```

```
## + wexp  1 1336.4
```

```
## + fin   1 1336.5
```

```
## <none>   1337.5
```

```
## + mar    1 1337.5
## + paro   1 1338.6
##
## Step:  AIC=1329.08
## Surv(week, arrest) ~ age + prio
##
##          Df      AIC
## + fin    1 1327.7
## + mar    1 1329.0
## <none>    1329.1
## + wexp   1 1330.0
## + paro   1 1330.9
##
## Step:  AIC=1327.71
## Surv(week, arrest) ~ age + prio + fin
##
##          Df      AIC
## + mar    1 1327.3
## <none>    1327.7
## + wexp   1 1328.6
## + paro   1 1329.4
##
## Step:  AIC=1327.35
## Surv(week, arrest) ~ age + prio + fin + mar
##
##          Df      AIC
## <none>    1327.3
## + wexp   1 1328.8
## + paro   1 1329.2
```

```
summary(for.model)
```

```
## Call:
## coxph(formula = Surv(week, arrest) ~ age + prio + fin + mar,
##       data = recid)
##
## n= 432, number of events= 114
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## age  -0.06042    0.94137  0.02085 -2.897  0.00376 **
## prio   0.09751    1.10243  0.02722  3.583  0.00034 ***
## fin  -0.36020    0.69753  0.19049 -1.891  0.05864 .
## mar  -0.53312    0.58677  0.37276 -1.430  0.15266
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## age      0.9414      1.0623      0.9037      0.9806
## prio      1.1024      0.9071      1.0452      1.1628
## fin      0.6975      1.4336      0.4802      1.0132
## mar      0.5868      1.7042      0.2826      1.2183
##
## Concordance= 0.633 (se = 0.027 )
## Likelihood ratio test= 31.41 on 4 df,  p=3e-06
## Wald test              = 29.98 on 4 df,  p=5e-06
## Score (logrank) test = 31.25 on 4 df,  p=3e-06
```

```
# Estimated Survival Curves #
```

```
newdata <- data.frame(fin = c(1, 0), age = 30, wexp = c(1, 0),  
                      mar = 0, paro = 0, prio = c(0, 4))  
  
ggsurvplot(survfit(recid.ph, newdata), data = newdata, break.y.by = 0.1,  
           palette = c("purple", "black"), ylab = "survival probability",  
           xlab = "week", legend.labs = c("1", "2"), legend.title = "subject")
```

```
## Warning: `gather_()` was deprecated in tidyr 1.2.0.
```

```
## i Please use `gather()` instead.
```

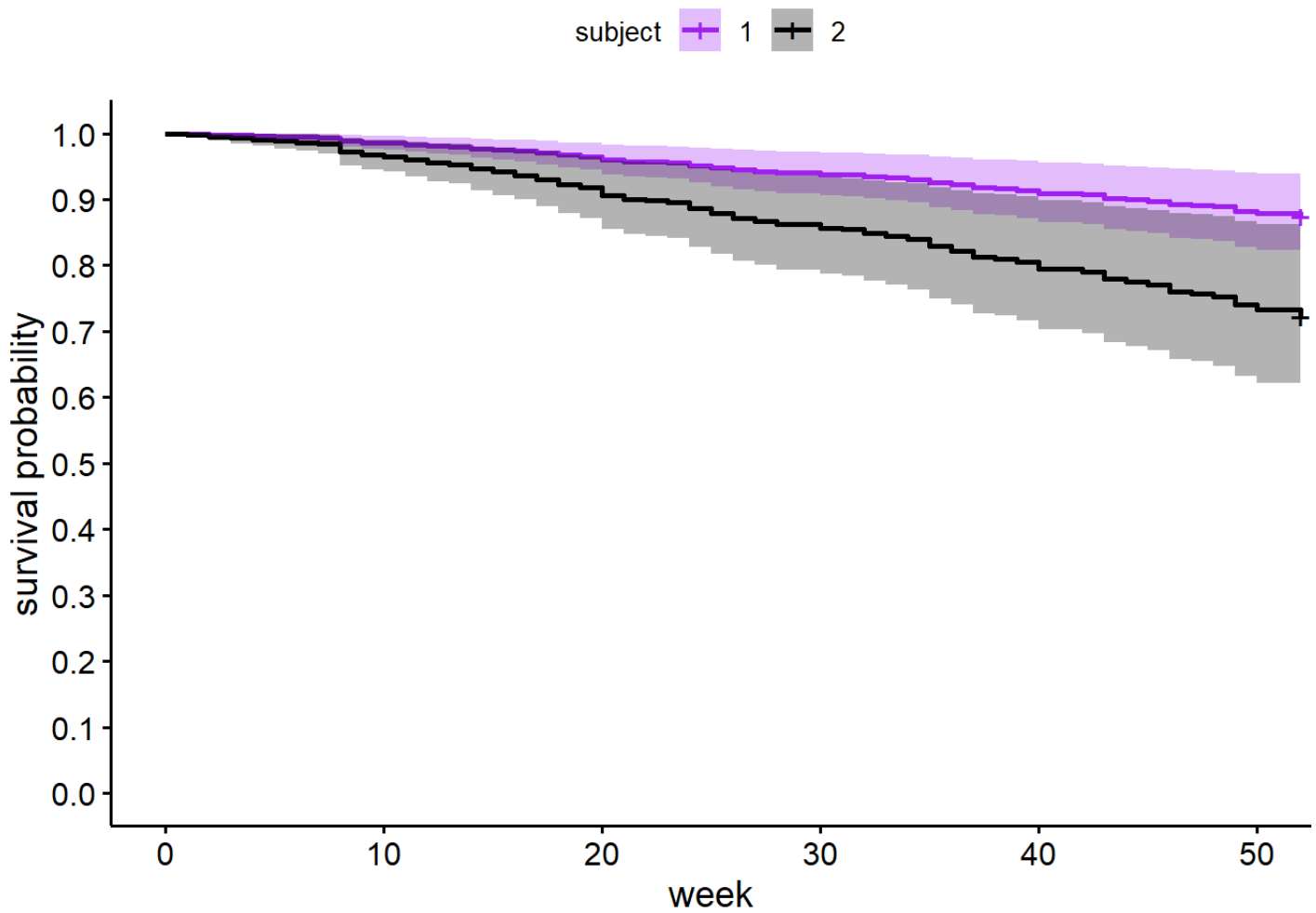
```
## i The deprecated feature was likely used in the survminer package.
```

```
## Please report the issue at <https://github.com/kassambara/survminer/issues>.
```

```
## This warning is displayed once every 8 hours.
```

```
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
```

```
## generated.
```



4.0.1 Python code for Cox regression

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from lifelines import CoxPHFitter

recid = r.recid

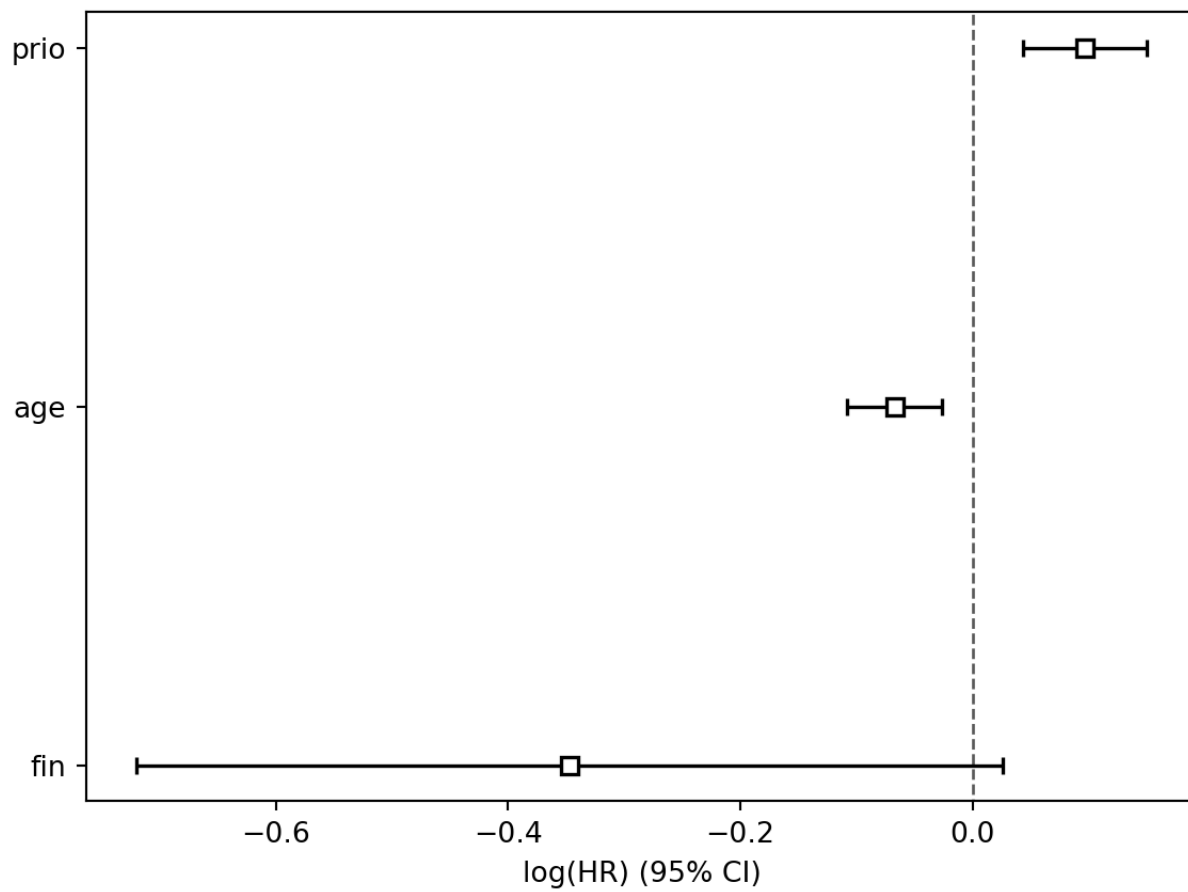
recid_ph=recid[['fin','age', 'prio','arrest','week']]
cph = CoxPHFitter()
cph.fit(recid_ph, duration_col='week', event_col='arrest')

## <lifelines.CoxPHFitter: fitted with 432 total observations, 318 right-censored

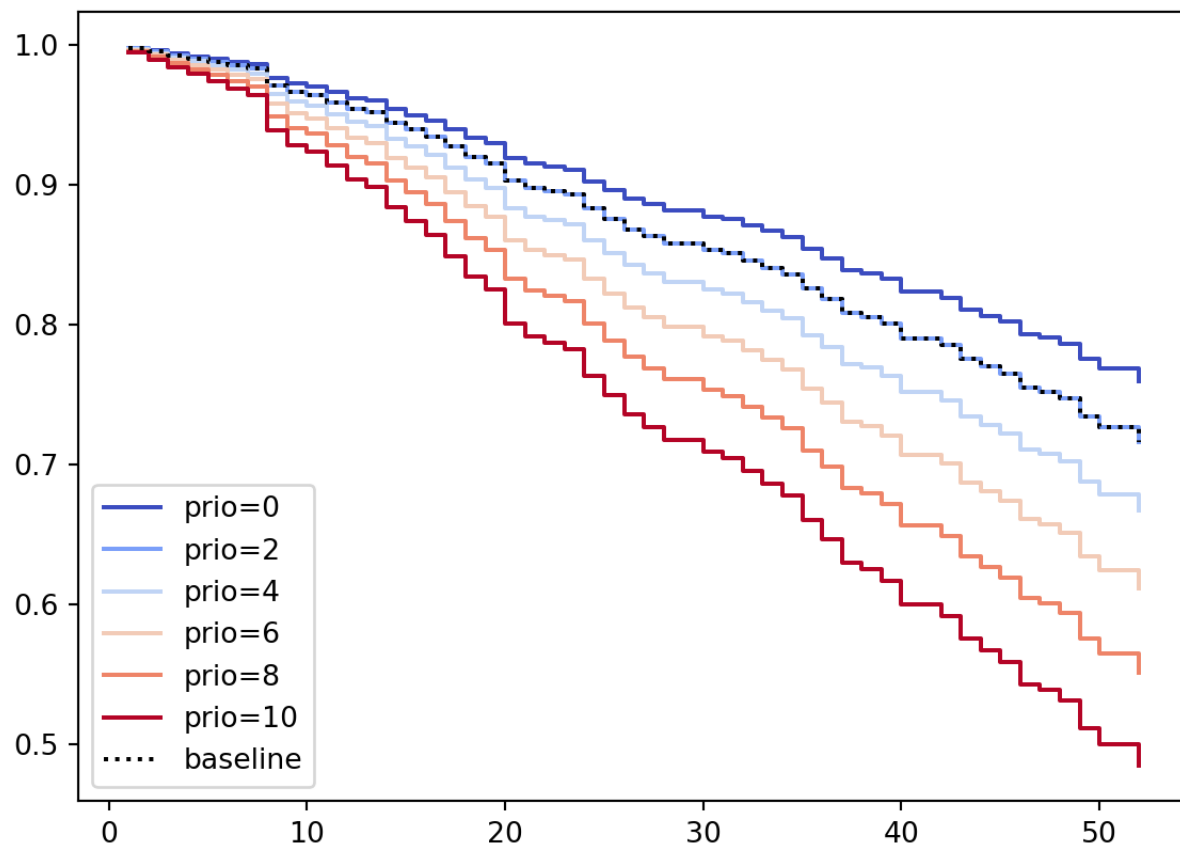
cph.print_summary()
```

```
## <lifelines.CoxPHFitter: fitted with 432 total observations, 318 right-censored
##           duration col = 'week'
##           event col = 'arrest'
##           baseline estimation = breslow
##           number of observations = 432
##           number of events observed = 114
##           partial log-likelihood = -660.86
##           time fit was run = 2024-11-04 15:34:08 UTC
##
## ---
##           coef exp(coef)  se(coef)  coef lower 95%  coef upper 95% exp(coef)
## covariate
## fin      -0.35      0.71      0.19           -0.72           0.03
## age      -0.07      0.94      0.02           -0.11          -0.03
## prio       0.10      1.10      0.03           0.04           0.15
##
##           cmp to      z      p  -log2(p)
## covariate
## fin           0.00 -1.82   0.07      3.87
## age           0.00 -3.22 <0.005     9.60
## prio           0.00  3.56 <0.005    11.37
##
## ---
## Concordance = 0.63
## Partial AIC = 1327.71
## log-likelihood ratio test = 29.05 on 3 df
## -log2(p) of ll-ratio test = 18.80
```

```
cph.plot()
```



```
cph.plot_partial_effects_on_outcome(covariates='prio', values=[0, 2, 4, 6, 8, 10])
```



4.1 Diagnostics

Assumptions made for the Cox PH model is linearity for the continuous variables and proportional hazards throughout time (i.e. at EACH time point, the proportion between the hazards remains the same!). We can assess both of these through use of residuals.

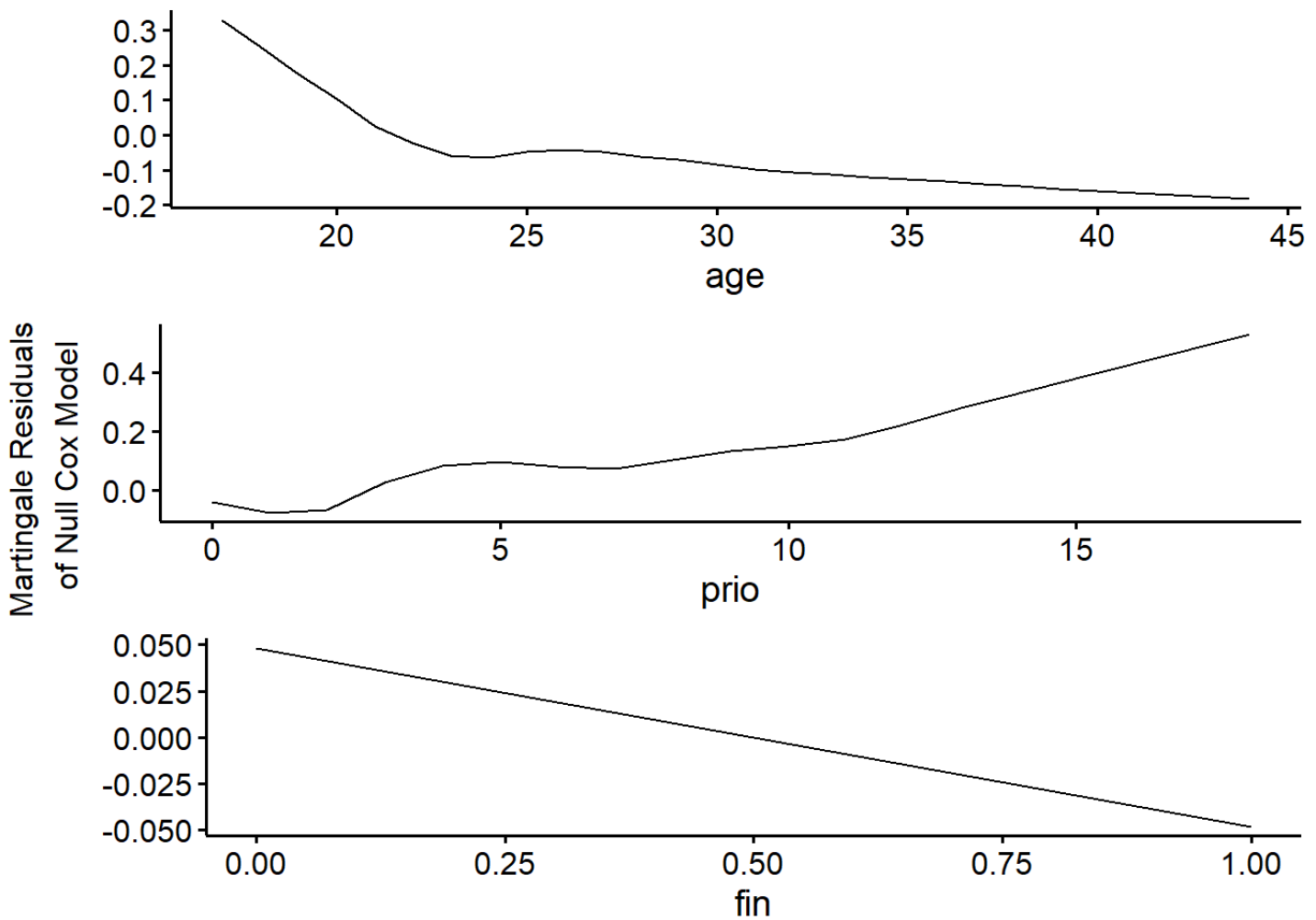
Concordance

```
recid.ph = coxph(Surv(week, arrest) ~ fin + age + prio, data = recid)
concordance(recid.ph)
```

```
## Call:
## concordance.coxph(object = recid.ph)
##
## n= 432
## Concordance= 0.6302 se= 0.02729
## concordant discordant      tied.x      tied.y      tied.xy
##      26736      15651      195      111      0

## Martingale residuals...Linearity

recid.lin <- coxph(Surv(week, arrest) ~ age + prio + fin, data = recid)
survminer::ggcoxfunctional(recid.lin, data=recid)
```



Try transformations or can bin

```
recid1<-recid %>% mutate(agebin = case_when(
  age < 20 ~ 0,
  age < 30 ~ 1,
  age >= 30 ~2))
```

```
recid.ph1 = coxph(Surv(week, arrest) ~ fin + factor(agebin) + prio, data = recid1)
```

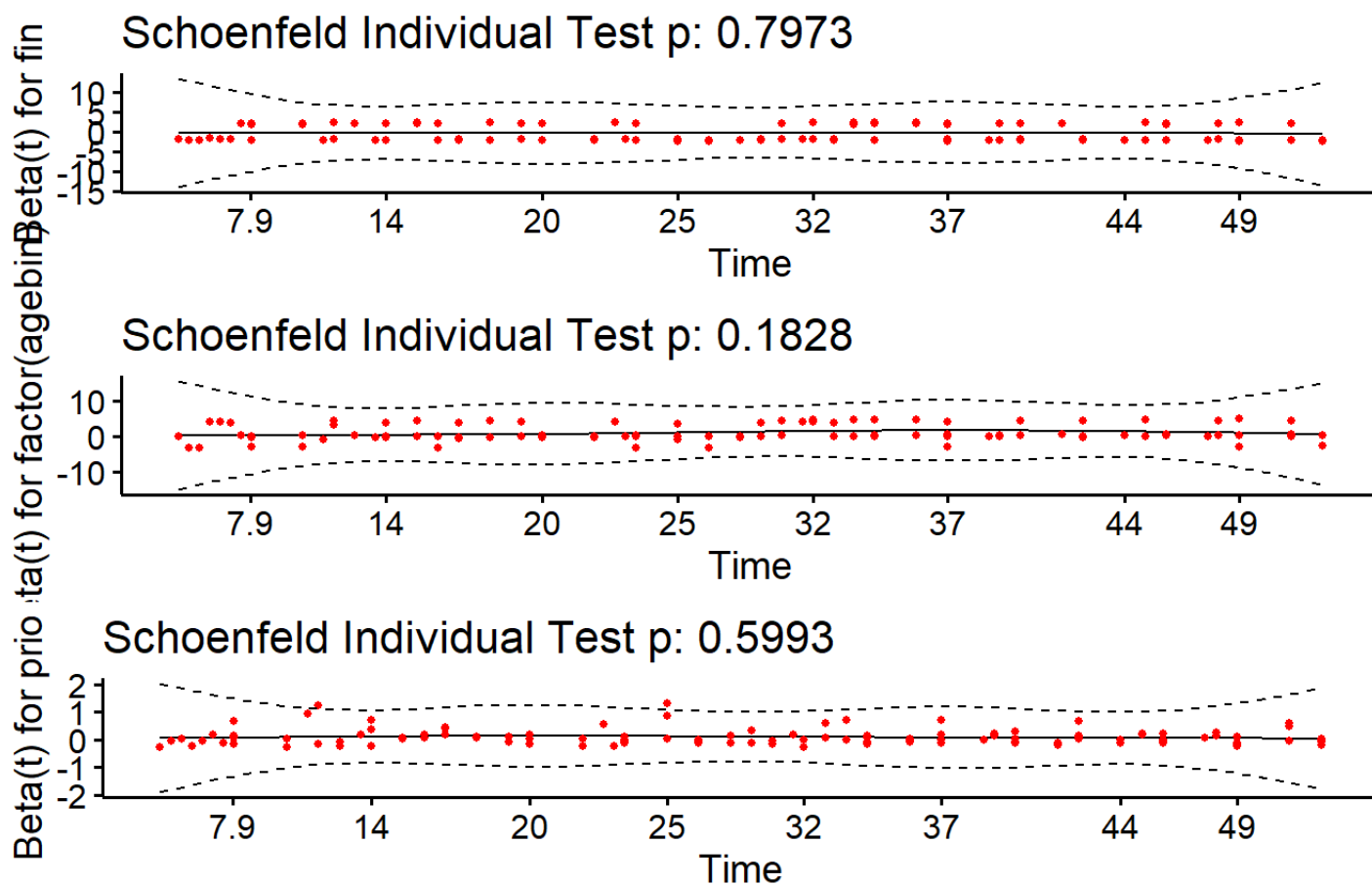
Proportional Hazard Test – Schoenfeld Residuals

```
recid.ph.zph <- cox.zph(recid.ph1)
recid.ph.zph
```

```
##               chisq df    p
## fin           0.066  1 0.80
## factor(agebin) 3.399  2 0.18
## prio          0.276  1 0.60
## GLOBAL        3.880  4 0.42
```

```
ggcoxzph(recid.ph.zph)
```

Global Schoenfeld Test p: 0.4225



To fit time dependent coefficients:

Time-Dependent Coefficients

```

recid.ph.tdc <- coxph(Surv(week, arrest) ~ fin + wexp + mar + paro + prio+ age +
                      tt = function(x, time, ...){x*log(time)})
summary(recid.ph.tdc)

```

```
## Call:
```

```
## coxph(formula = Surv(week, arrest) ~ fin + wexp + mar + paro +
##      prio + age + tt(age), data = recid, tt = function(x, time,
##      ...) {
##      x * log(time)
## })
```

```
##
```

```
##      n= 432, number of events= 114
```

```
##
```

```
##              coef exp(coef) se(coef)      z Pr(>|z|)
## fin          -0.36527   0.69401  0.19087 -1.914  0.05566 .
## wexp         -0.13317   0.87531  0.21247 -0.627  0.53080
## mar          -0.45279   0.63585  0.38041 -1.190  0.23394
## paro         -0.08490   0.91860  0.19534 -0.435  0.66382
## prio          0.09177   1.09611  0.02880  3.186  0.00144 **
## age           0.12174   1.12946  0.06535  1.863  0.06249 .
## tt(age)     -0.05931   0.94242  0.02182 -2.718  0.00658 **
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
##              exp(coef) exp(-coef) lower .95 upper .95
## fin              0.6940      1.4409    0.4774    1.0089
## wexp              0.8753      1.1424    0.5772    1.3275
## mar              0.6358      1.5727    0.3017    1.3402
## paro              0.9186      1.0886    0.6264    1.3471
```

```
## prio      1.0961      0.9123      1.0359      1.1598
## age       1.1295      0.8854      0.9937      1.2838
## tt(age)   0.9424      1.0611      0.9030      0.9836
##
## Concordance= 0.648 (se = 0.027 )
## Likelihood ratio test= 38.72 on 7 df, p=2e-06
## Wald test          = 35.09 on 7 df, p=1e-05
## Score (logrank) test = 36.97 on 7 df, p=5e-06
```

```
recid.ph.tdc <- coxph(Surv(week, arrest) ~ fin + prio + age + tt(age), data = I
  tt = function(x, time, ...){x*log(time)})
summary(recid.ph.tdc)
```

```
## Call:
## coxph(formula = Surv(week, arrest) ~ fin + prio + age + tt(age),
##       data = recid, tt = function(x, time, ...) {
##         x * log(time)
##       })
##
## n= 432, number of events= 114
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## fin          -0.34878   0.70555  0.19025 -1.833 0.066762 .
## prio           0.09837   1.10337  0.02731  3.603 0.000315 ***
## age           0.11991   1.12739  0.06623  1.810 0.070220 .
## tt(age)      -0.06195   0.93993  0.02207 -2.807 0.005005 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## fin              0.7055      1.4173    0.4859    1.0244
## prio             1.1034      0.9063    1.0459    1.1640
## age              1.1274      0.8870    0.9901    1.2837
## tt(age)          0.9399      1.0639    0.9001    0.9815
##
## Concordance= 0.635 (se = 0.028 )
## Likelihood ratio test= 36  on 4 df,   p=3e-07
## Wald test              = 32.41  on 4 df,   p=2e-06
## Score (logrank) test = 33.75  on 4 df,   p=8e-07
```

The following code is for time-varying variables (values of the variable that change over time). Notice that we are using a different data set (needs to be structured differently).

Time Varying Variables

```
recid_long.ph <- coxph(Surv(start, stop, arrested) ~ fin + age + prio + employed,
summary(recid_long.ph)
```

```
## Call:
```

```
## coxph(formula = Surv(start, stop, arrested) ~ fin + age + prio +
##     employed, data = recid_long)
```

```
##
```

```
##   n= 19809, number of events= 114
```

```
##
```

```
##           coef exp(coef) se(coef)      z Pr(>|z|)
## fin      -0.33051   0.71856  0.19012 -1.738  0.08214 .
## age      -0.04977   0.95145  0.02053 -2.424  0.01537 *
## prio       0.08364   1.08724  0.02775  3.014  0.00258 **
## employed -1.34815   0.25972  0.24928 -5.408 6.37e-08 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
##           exp(coef) exp(-coef) lower .95 upper .95
## fin           0.7186      1.3917   0.4950   1.0430
## age           0.9515      1.0510   0.9139   0.9905
## prio          1.0872      0.9198   1.0297   1.1480
## employed      0.2597      3.8503   0.1593   0.4233
```

```
##
```

```
## Concordance= 0.703 (se = 0.023 )
```

```
## Likelihood ratio test= 66.19 on 4 df, p=1e-13
```

```
## Wald test           = 53.93 on 4 df, p=5e-11
```

```
## Score (logrank) test = 62.05 on 4 df, p=1e-12
```

```
recid_lag.ph <- coxph(Surv(start, stop, arrested) ~ fin + age + prio + employed,
summary(recid_lag.ph)
```

```
## Call:
```

```
## coxph(formula = Surv(start, stop, arrested) ~ fin + age + prio +
##      employed, data = recid_lag)
```

```
##
```

```
##      n= 19377, number of events= 113
```

```
##
```

```
##              coef exp(coef) se(coef)      z Pr(>|z|)
## fin          -0.32464    0.72278  0.19077 -1.702 0.088803 .
## age          -0.05462    0.94685  0.02073 -2.635 0.008409 **
## prio           0.09158    1.09590  0.02753  3.326 0.000880 ***
## employed    -0.81123    0.44431  0.21648 -3.747 0.000179 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
##              exp(coef) exp(-coef) lower .95 upper .95
## fin              0.7228      1.3835    0.4973    1.0505
## age              0.9468      1.0561    0.9092    0.9861
## prio             1.0959      0.9125    1.0383    1.1567
## employed         0.4443      2.2507    0.2907    0.6791
```

```
##
```

```
## Concordance= 0.665 (se = 0.027 )
```

```
## Likelihood ratio test= 44.5 on 4 df, p=5e-09
```

```
## Wald test          = 40.91 on 4 df, p=3e-08
```

```
## Score (logrank) test = 43.68 on 4 df, p=7e-09
```

4.1.1 Python for Diagnostics

For testing proportional hazard assumption:

```
cph.check_assumptions(recid_ph, p_value_threshold = 0.05)
```

The ``p\_value\_threshold`` is set at 0.05. Even under the null hypothesis of no
 ## covariates will be below the threshold by chance. This is compounded when there
 ## Similarly, when there are lots of observations, even minor deviances from the
 ## assumption will be flagged.

##

With that in mind, it's best to use a combination of statistical tests and visual
 ## the most serious violations. Produce visual plots using ``check_assumptions(...
 ## and looking for non-constant lines. See link [A] below for a full example.

##

```
## <lifelines.StatisticalResult: proportional_hazard_test>
## null_distribution = chi squared
## degrees_of_freedom = 1
##          model = <lifelines.CoxPHFitter: fitted with 432 total observations>
##          test_name = proportional_hazard_test
##
## ---
##          test_statistic      p  -log2(p)
## age    km                6.29 0.01      6.37
##          rank              6.62 0.01      6.63
## fin    km                0.00 1.00      0.01
##          rank              0.00 0.98      0.03
## prio   km                0.82 0.37      1.45
##          rank              0.77 0.38      1.40
##
##
## 1. Variable 'age' failed the non-proportional test: p-value is 0.0101.
```

```
##
## Advice 1: the functional form of the variable 'age' might be incorrect. The
## non-linear terms missing. The proportional hazard test used is very sensitive
## functional forms. See documentation in link [D] below on how to specify a func
##
## Advice 2: try binning the variable 'age' using pd.cut, and then specify it
## ...]'` in the call in `.fit`. See documentation in link [B] below.
##
## Advice 3: try adding an interaction term with your time variable. See docum
## below.
##
##
## ---
## [A] https://lifelines.readthedocs.io/en/latest/jupyter\_notebooks/Proportional
## [B] https://lifelines.readthedocs.io/en/latest/jupyter\_notebooks/Proportional
## [C] https://lifelines.readthedocs.io/en/latest/jupyter\_notebooks/Proportional
## [D] https://lifelines.readthedocs.io/en/latest/jupyter\_notebooks/Proportional
## [E] https://lifelines.readthedocs.io/en/latest/jupyter\_notebooks/Proportional
##
## []
```

Python cannot program time-varying coefficients. However, it is capable of doing time-varying variables:

```
from lifelines import CoxTimeVaryingFitter

recid_1=r.recid_long
recid_long=recid_1[['fin', 'age', 'prio', 'employed','start','stop','subj','arrested']]
ctv = CoxTimeVaryingFitter()
ctv.fit(recid_long, id_col="subj", event_col="arrested", start_col="start", stop_col="stop")
```

```
## Iteration 1: norm_delta = 6.45e-01, step_size = 0.9500, log_lik = -675.38063,  
## Iteration 2: norm_delta = 1.68e-01, step_size = 0.9500, log_lik = -643.80170,  
## Iteration 3: norm_delta = 2.27e-02, step_size = 0.9500, log_lik = -642.30569,  
## Iteration 4: norm_delta = 1.48e-03, step_size = 1.0000, log_lik = -642.28397,  
## Iteration 5: norm_delta = 1.14e-06, step_size = 1.0000, log_lik = -642.28389,  
## Convergence completed after 5 iterations.  
## <lifelines.CoxTimeVaryingFitter: fitted with 19809 periods, 432 subjects, 114
```

```
ctv.print_summary()
```

```
## <lifelines.CoxTimeVaryingFitter: fitted with 19809 periods, 432 subjects, 114
##           event col = 'arrested'
## number of subjects = 432
##  number of periods = 19809
##  number of events = 114
## partial log-likelihood = -642.28
##  time fit was run = 2024-11-04 15:34:11 UTC
##
## ---
##           coef exp(coef)  se(coef)  coef lower 95%  coef upper 95% exp(coef)
## covariate
## fin          -0.33      0.72      0.19          -0.70          0.04
## age          -0.05      0.95      0.02          -0.09          -0.01
## prio           0.08      1.09      0.03           0.03          0.14
## employed     -1.35      0.26      0.25          -1.84          -0.86
##
##           cmp to      z      p  -log2(p)
## covariate
## fin           0.00 -1.74  0.08      3.61
## age           0.00 -2.42  0.02      6.02
## prio           0.00  3.01 <0.005      8.60
## employed       0.00 -5.41 <0.005     23.90
## ---
## Partial AIC = 1292.57
## log-likelihood ratio test = 66.19 on 4 df
## -log2(p) of ll-ratio test = 42.66
```